

Business Insight Report

Retail Inventory Risk & Demand Forecasting

Prepared by Amrutha K | Dataset: Retail Store Inventory (Kaggle, 73,100 rows) | Tools: Python, Power BI

Executive Summary

This report analyses two years of daily inventory and sales data across 5 stores and 20 products to answer two operational questions: which products are at risk of stocking out, and how far ahead can demand be reliably forecast? The analysis finds that inventory risk is a widespread, evenly-distributed problem rather than a localized one — nearly half of all store-product combinations currently sit below a data-driven reorder point. It also finds that demand forecasting is reliable at a weekly level but not at a daily level, with clear implications for how replenishment decisions should be timed.

Metric	Value	Insight
Total Records Analysed	73,100 rows	5 stores × 20 products, 2 years daily data
Store-Product Pairs Needing Reorder	47%	Nearly half of inventory is understocked
Weekly Forecast Accuracy	26% MAE	Reliable enough for planning decisions
Daily Forecast Accuracy	65% MAE	Too noisy for day-level decisions

Methodology

Data was explored and cleaned in Python (pandas), including data quality checks, feature engineering (sell-through rate, 7-day rolling demand, stock cover days), and hypothesis testing. Reorder points were calculated using a standard safety-stock formula (average demand × lead time + safety stock), with lead time and service level assumptions revised after checking them against observed inventory scale. Demand forecasting used Exponential Smoothing (Holt-Winters), validated against actual sales using Mean Absolute Error at both daily and weekly aggregation levels. Findings were visualized in a three-page interactive Power BI dashboard.

Key Findings & Recommendations

Finding 1: Inventory Risk Is Widespread, Not Localized

47% of store-product combinations currently sit below their calculated reorder point. Critically, this risk is evenly distributed across all categories and regions (roughly 24-27% low-stock rate everywhere) — there is no single category, region, or store driving the problem. This means the issue is systemic (e.g., a replenishment process gap) rather than isolated to one product line.

Recommendation: Implement an automated reorder alert system rather than manual, category-by-category stock reviews, since the risk cuts evenly across the entire catalogue.

Finding 2: Weekly Forecasts Are Reliable; Daily Forecasts Are Not

An Exponential Smoothing model produced a Mean Absolute Error of 65% of average sales at the daily level, but only 26% at the weekly level. Daily demand is inherently noisy at the individual SKU level; aggregating to weekly totals cancels out much of that random variation.

Recommendation: *Base replenishment and planning decisions on weekly demand forecasts, not daily ones. Daily level forecasts should only be used for short-term operational awareness, not for ordering decisions.*

Finding 3: Discounting Does Not Drive Sales Volume in This Business

Discount level showed effectively no relationship with units sold ($r = 0.003$). This is a notably different pattern from a separate e-commerce analysis, where discounting had a severe negative effect on profit margin. The two findings are not contradictory — they simply show that the impact of discounting is business- and context-specific, not a universal rule.

Recommendation: *Avoid assuming promotional/discount strategies from one business will transfer to another; validate discount impact per business context before acting on it.*

Finding 4: Holiday and Weather Effects Are Small and Not Actionable Alone

Holiday/Promotion periods showed no meaningful lift in sales (136.5 vs 136.4 average units). Weather showed a statistically significant effect ($p = 0.012$) but the practical size of the effect was only around 2% — too small to justify operational changes on its own.

Recommendation: *Do not treat weather or holiday flags as standalone planning signals; if used at all, they should be minor adjustments layered on top of the core weekly demand forecast, not primary drivers.*

Conclusion

The clearest, most actionable finding in this analysis is that inventory risk is systemic and forecasting reliability depends heavily on the time horizon used. Rather than reacting to stockouts store-by-store, the business would benefit most from a standardized, automated reorder process built on weekly (not daily) demand forecasts — addressing the root cause affecting nearly half of all inventory positions, rather than treating each stockout as an isolated incident.